Complex Homework #7

1. Suppose $f : \Delta^* \to \mathbb{C}$ is holomorphic, and

$$|f(z)| \leq 1/\sqrt{|z|}$$

for all $z \in \Delta$. Show that f extends to a holomorphic function $\Delta \to \mathbb{C}$.

Cosider the following limit, we have

$$\begin{aligned}
\lim_{z\to 0} |zf(z)| &= \lim_{z\to 0} |z|\,|f(z)| \\
&\leq \lim_{z\to 0} |z| \frac{1}{\sqrt{|z|}} \\
&= \lim_{z\to 0} \sqrt{|z|} \\
&= 0,
\end{aligned}$$

and so $\lim_{z\to 0} |zf(z)| = 0$. I particular we then have that $\lim_{z\to 0} zf(z) = 0$ since 0 is the only complex number with zero modulus.

Now, we can construct a holomorphic extension of f on Δ. (Note that this is the proof of an alternative characterization of the Riemann Removable singularity theorem that we saw in class).

Let $h : \Delta \to \mathbb{C}$ be defined by

$$h(z) = \begin{cases} z^2 f(z) & z \neq 0 \\ 0 & z = 0. \end{cases}$$

Notice that h is holomorphic on $\Delta^\star$ as the product of two holomorphic functions on $\Delta^\star$. We show that h is also holomorphic at $z = 0$. We have

$$h'(0) = \lim_{z\to 0} \frac{z^2 f(z) - 0}{z} = \lim_{z\to 0} zf(z) = 0,$$

where the last equality follows by the argument above. And so h is holomorphic at 0 and in particular $h'(0) = 0$.

Since h is holomorphic at $z = 0$ it has a power series expansion about $z = 0$

$$h(z) = c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots$$

However, by definition $c_0 = h(0) = 0$ and $c_1 = h'(0) = 0$ and so we actually have

$$h(z) = c_2 z^2 + c_3 z^3 + \cdots$$

Recall that for $z \neq 0$ we have

$$f(z) = \frac{h(z)}{z^2} = c_2 + c_3 z + c_4 z^2 + \cdots$$

In other words, the power series

$$c_2 + c_3 z + c_4 z^2 + \cdots$$

agrees with $f(z)$ for all $z \neq a$ in $\Delta^\star$. However, this power series ia also manifestly holomorphic on all of Δ. And so we have found a holomorphic function which extends f to $\Delta \to \mathbb{C}$.

$\blacksquare$

2. Show that if $f : \mathbb{C} \to \mathbb{C}$ is holomorphic and *injective*, then $f(z) = az + b$ for some $a, b \in \mathbb{C}$. (Hint: consider the behavior at 0 of $f(1/z)$).

Let $g(z) = f(1/z)$. Let us understand what kind of singularity occurs at $g(0)$. If g had a removable singularity at 0 then we would have that g has a power series expansion about $z = 0$ as $g(z) = c_0 + c_1 z + c_2 z + \cdots$ and then in particular $g(0) = c_0$ some finite value. However, this then implies that $f(z)$ is bounded for large z. And then, as an entire function, this would imply that f is constant. In other words $f(z) = b$ for some $b \in \mathbb{C}$, a special case of the desired form.

Now suppose f is non-constant. We cannot have a removable singularity for g at $z = 0$. Consider what happens if g has an essential singularity at $z = 0$. In this case we can apply the Casorati-Weierstrass theorem; we have that the image under g of any neighbourhood of the origin $D_\varepsilon(0)$ is dense in $\mathbb{C}$. By definition $g(D_\varepsilon(0)) = f(\mathbb{C} \setminus D_{1/\varepsilon}(0))$ is dense in $\mathbb{C}$. Now, recall that f is holomorphic and so is, in particular, an open map. That is the image of some other disc $f(D_\delta(0))$ is open in $\mathbb{C}$ and so must intersect $f(\mathbb{C} \setminus D_{1/\varepsilon}(0))$. However, for sufficiently small δ, ε we have that $D_\delta(0)$ and $\mathbb{C} \setminus D_{1/\varepsilon}(0)$ are disjoint, yet their images under f intersect. This means there are two distinct $z_1, z_2 \in \mathbb{C}$ with $f(z_1) = f(z_2)$, contradicting the injectivity of f. And so g cannot have an essential singularity at $z = 0$.

If f is nonconstant, we are left with a pole at $z = 0$ for g. That is the Laurent series for g at $z = 0$ is given by

$$g(z) = \frac{a_{-k}}{z^k} + \frac{a_{-k+1}}{z^{k-1}} + \cdots + a_0 + a_1 z + a_2 z^2 + \cdots$$

for some $a_i \in \mathbb{C}$ and with not all of the $a_i = 0$ for $i < 0$. Now since $g(z) = f(1/z)$ we have a Laurent series for $f(z)$ at $z = 0$ by plugging in $1/z$ into the series above. That is, near $z = 0$ we have

$$f(z) = a_{-k} z^k + a_{-k+1} z^{k-1} + \cdots a_{-1} z + a_0 + \frac{a_1}{z} + \frac{a_2}{z^2 k} + \cdots$$

However, we have that f is entire, and so is in particular not singular at $z = 0$. Thus we

have $a_i = 0$ for all $i \geq 1$. That is,

$$f(z) = a_0 + a_{-1}z + \cdots a_{-k}z^k.$$

Moreover, since f is entire this series holds everywhere on $\mathbb{C}$ (has infinite radius of convergence). I think I see this intuitively, but note to self to try and think about this using Hadamard

We have shown that f is a polynomial, but recall that $\mathbb{C}$ is algebraically closed. And so if f is degree k then it has k roots in $\mathbb{C}$. That is, there are k points in $\mathbb{C}$ where $f(z) = 0$. However, it must be that $k = 1$, otherwise we would have shown that f is not injective. That is, we in fact have that f is a degree 1 polynomial

$$f(z) = a_0 + a_{-1}z,$$

for some $a_0, a_1 \in \mathbb{C}$.

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3. Find the residue of $\sin^3(1/z)$ at 0.

Recall that the residue of a meromorphic function $f : U \to \mathbb{C}$ at a point $p \in U$ is given by the a_{-1} coefficient in it's Laurent expansion about p. For this question we determine the Laurent expansion of $\sin^3(z)$ about $z = 0$ by using power series that we already know.

Recall the Taylor expansion of $\sin z$ about $z = 0$ is given by

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$$

And so in particular we have the following Laurent series expansion of $\sin^3(1/z)$ about $z = 0$

$$\sin^3\left(\frac{1}{z}\right) = \left[\frac{1}{z} - \frac{1}{3!}\frac{1}{z^3} + \frac{1}{5!}\frac{1}{z^5} - \cdots\right]^3$$

Now, notice that the largest power of z in this expansion is going to be $1/z^3$ with all other terms having a lower power of z. That is the coefficients of the Laurent series expansion is $a_n = 0$ for all $n \geq -2$. In particular $a_{-1} = 0$ and so the residue of $\sin^3(1/z)$ about $z = 0$ is 0.

Note that althought the residue of our function is 0 at $z = 0$, our function does have a singularity at 0. In fact, the Laurent series expansion about $z = 0$ has infinitely many terms with negative powers of z and so the singularity at $z = 0$ is essential.

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4. (a) Let $U \subset \mathbb{C}$ open, and let $f, g : U \to \mathbb{C}$ holomorphic. Show that

$$Res(gf'/f, p) = g(p)Mult_p(f)$$

(where $Res(\cdot, p)$ is the residue at p, and the left hand side is the multiplicity of vanishing of f at p). What does this give for $g(z) = z$ when we apply the residue formula?

(b) Let

$$p_u(z) = z^n + a_{n-1}(u)z^{n-1} + \cdots + a_0(u)$$

be a holomorphic *family* of polynomials, where each $a_i : \Delta \to \mathbb{C}$ is a holomorphic function. Suppose $p_0(z)$ has all roots simple (multiplicity 1). Show that there is some disc D containing 0 and holomorphic functions $r_i : D \to \mathbb{C}$ such that

$$p_u(z) = (z - r_1(u)) \cdots (z - r_n(u))$$

for all $u \in D$ and $z \in \Delta$.

Recall that $res_p(gf'/f)$ is the a_{-1} coefficient in the Laurent series expansion of gf'/f about p. We determine this expansion. Near p we can write

$$f(z) = (z - p)^k h(z)$$

where h is some holomorphic function on a neighbourhood about p which does not vanish on some neighbourhood about p (this latter claim follows from isolation of zeroes of holomorphic functions). Applying usual differentiation rules gives $f'(z) = k(z - p)^{k-1}h(z) + (z - p)^k h'(z)$. And so,

$$\frac{f'(z)}{f(z)} = \frac{k}{z - p} + \frac{h'(z)}{h(z)}$$

where the latter term is holomorphic about p, since there exists a neighbourhood about p where $h(z)$ is non-vanishing. As an aside, note then that we have just shown that $res_p(f'/f) = k = Mult_p(f)$.

Now consider

$$\frac{gf'}{f} = g(z)\left[\frac{k}{z - p} + \frac{h'(z)}{h(z)},\right].$$

Since g is holomorphic on U it has a power series expansion about p, say $g(z) = b_0 +$

$b_1(z-p) + b_2(z-p)^2 + \cdots$. It follows then that

$$res_p(gf'/f) = \text{“}a_{-1}\left(g(z)\left[\frac{k}{z-p} + \frac{h'(z)}{h(z)}\right]\right)'' = b_0 k = g(p)Mult_p(f).$$

If in particular $g(z) = z$ then we have $res_p(zf'/f) = pMult_p(f)$. And so the Residue formula gives

$$\int_{\partial D} \frac{zf'(z)}{f(z)} dz = 2\pi i \sum_{p \in D} pMult_p(f).$$

A weighted sum of points in an (appropriate) disc, weighted by the order of vanishing of f at those points. Note that only finitely many terms of the sum on the right hand side of this expression are non-zero.

Indeed, we have seen that the zeroes of a holomorphic function are isolated and so form a discrete set inside $\bar{D}$; a closed discrete set contained in a compact set must be finite. (wait actually it's unclear to me whether the collection of zeroes inside a closed disc is without assuming that it's first finite, since it's a union of points). And then if f has only finitely many zeroes in $\bar{D}$ then it must have only finitely many zeroes in D.

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